

Douwe Rijs

Robotics & Machine Learning Engineer · MSc Robotics, TU Delft

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Robotics MSc student at TU Delft with a Mechanical Engineering background and a strong focus on machine learning, perception and control systems. Hands-on R&D experience in computer vision, transformer models and enterprise software development at Allseas. Motivated to apply advanced robotics and ML to real-world industrial systems.

EXPERIENCE

Student Engineer — Boeing Innovation Lab, Neu-Isenburg (Germany) Sep – Dec 2026 (upcoming)
MSc internship on the Autoturn project (automated aircraft turnaround): prototyping a common data infrastructure for ground-crew-to-aircraft communication, turnaround progress interfaces and exterior localisation tags, in an agile international team.

Assistant R&D Engineer — Allseas, Delft Feb 2025 – Aug 2026

- Designed and built a computer-vision/VLM pipeline for automated pipe-ID recognition — see Robotics & AI Projects below.
- Built a network observability platform monitoring 100+ switches via SNMP, aggregating port-level telemetry (data rate, VLAN, link status) for real-time diagnostics.
- Developed and maintained production software in Python, Go and React/TypeScript within an enterprise R&D environment; automated versioning and releases with Azure DevOps CI/CD.

EDUCATION

MSc Robotics — Delft University of Technology 2025 – present
Machine learning, deep reinforcement learning, machine perception (sensor fusion, Kalman filtering, LiDAR/radar/stereo vision), robot dynamics & control, motion planning (RRT*, MPC).

BSc Mechanical Engineering — Delft University of Technology 2022 – Oct 2025
Thesis (8.0/10): chemotherapy-injection assisting device — medical-device regulations & CE certification. Minor in Computer Science (30 EC, EWI): ML fundamentals, camera geometry & 3D reconstruction, algorithms & software engineering; team game-development final project.

VWO (pre-university) — Haarlemmermeerlyceum, Hoofddorp 2016 – 2022

ROBOTICS & AI PROJECTS

Autonomous Greenhouse Monitoring Robot & Digital Twin — TU Delft, team of 5

- Built an autonomous greenhouse robot in ROS 2 using Nav2 and SLAM Toolbox to map flower beds and stream plant, pest and climate data into a live digital twin.
- Developed the robot arm-control package and integrated SLAM, autonomous navigation, perception and state-machine components into a complete system; validated YOLOv8 detection and AprilTag localisation in simulation and on physical hardware.
- Managed the team's GitLab development workflow; delivered 7/8 client requirements within an 8-week development cycle.

Vision-Only Autonomous Drone Flight — TU Delft MAVLab

- Vision-only obstacle avoidance for a Parrot Bebop 2 flying autonomously in the CyberZoo using onboard compute only.
- Implemented border, gate and vertical-obstacle detection in C/C++ inside the Paparazzi autopilot; optimised the vision stack to 8–11 FPS on the drone's onboard ARM processor.

Pipe-ID Recognition from Steel Pipe Imagery — Allseas R&D

- Developed an end-to-end CV/VLM pipeline reading stencilled & chalked pipe IDs from photos taken inside pipeline pipes, increasing exact-ID accuracy from ~1% to ~80%.
- Benchmarked four OCR architectures (Tesseract, EasyOCR, PaddleOCR, TrOCR), diagnosed failure modes and pivoted to a tuned VLM approach after classical OCR plateaued.
- Improved exact-match accuracy from 29% to 59% through YOLO oriented-bounding-box crop preprocessing alone — demonstrating that preprocessing outweighed model choice.

Multi-Sensor 3D Object Detection for Autonomous Driving — TU Delft

- Built a LiDAR/camera/radar perception pipeline for 3D bounding-box detection on the real-world View-of-Delft driving dataset, handling noisy and sparse sensor data.

SKILLS

Programming	Python, Go, TypeScript/JavaScript, React, C++, MATLAB
Machine Learning	CNNs, RNNs, Transformers, VLMs, deep reinforcement learning, SVM, PCA, PyTorch, OpenCV, Hugging Face, YOLO
Robotics	ROS 2, SLAM, Nav2, sensor fusion, Kalman filtering, LiDAR, radar, stereo vision, pose estimation
Control & Planning	PID, MPC, impedance control, robot dynamics, RRT*, search algorithms
Tools & DevOps	Git, Azure DevOps, CI/CD, Docker, Linux, observability & monitoring
Languages	Dutch (native), English (fluent)